

Project Development Phase

Project Title: **Script Controlled AC L – Restrict Record Access Based on Field**

S.No	Team Role	Name	Register Number
1	Team Leader	Parimala	ASALUUAU020224128026
2	Team Members	Rajeswari	ASALUUAU020224128029
3		Priyanka	ASALUUAU020224128028
4		Rathika	ASALUUAU020224128030
5		Mounika	ASALUUAU020224128025

The development phase focuses on implementing Access Control Lists (ACLs) and server-side scripts to control access to records based on the value of a particular field.

1. Create the Required Table and Field

A table is created with a field that determines who should have access to the record.

Example:

1. Table: Project Request
2. Field: Access Level
3. Possible values:
 - Public
 - Team
 - Restricted

The field value becomes the basis for deciding whether a user can access the record.

2. Define User Roles:

Different roles are assigned to users according to their responsibilities

ROLE	DESCRIPTION	ACCESS
 ADMIN	Has full control over the system.	Can create, read, update, and delete all records.
 MANAGER	Can access and manage records based on team or department.	Can create and update permitted records, and read records based on access rules.
 EMPLOYEE	Can access only the records specifically allowed.	Can read allowed records. Limited create/update access.
 UNAUTHORIZED USER	No authorization to access restricted data.	Access denied to all protected records.

3. Configure ACL

An ACL (Access Control List) is created for the required table and operation, such as:

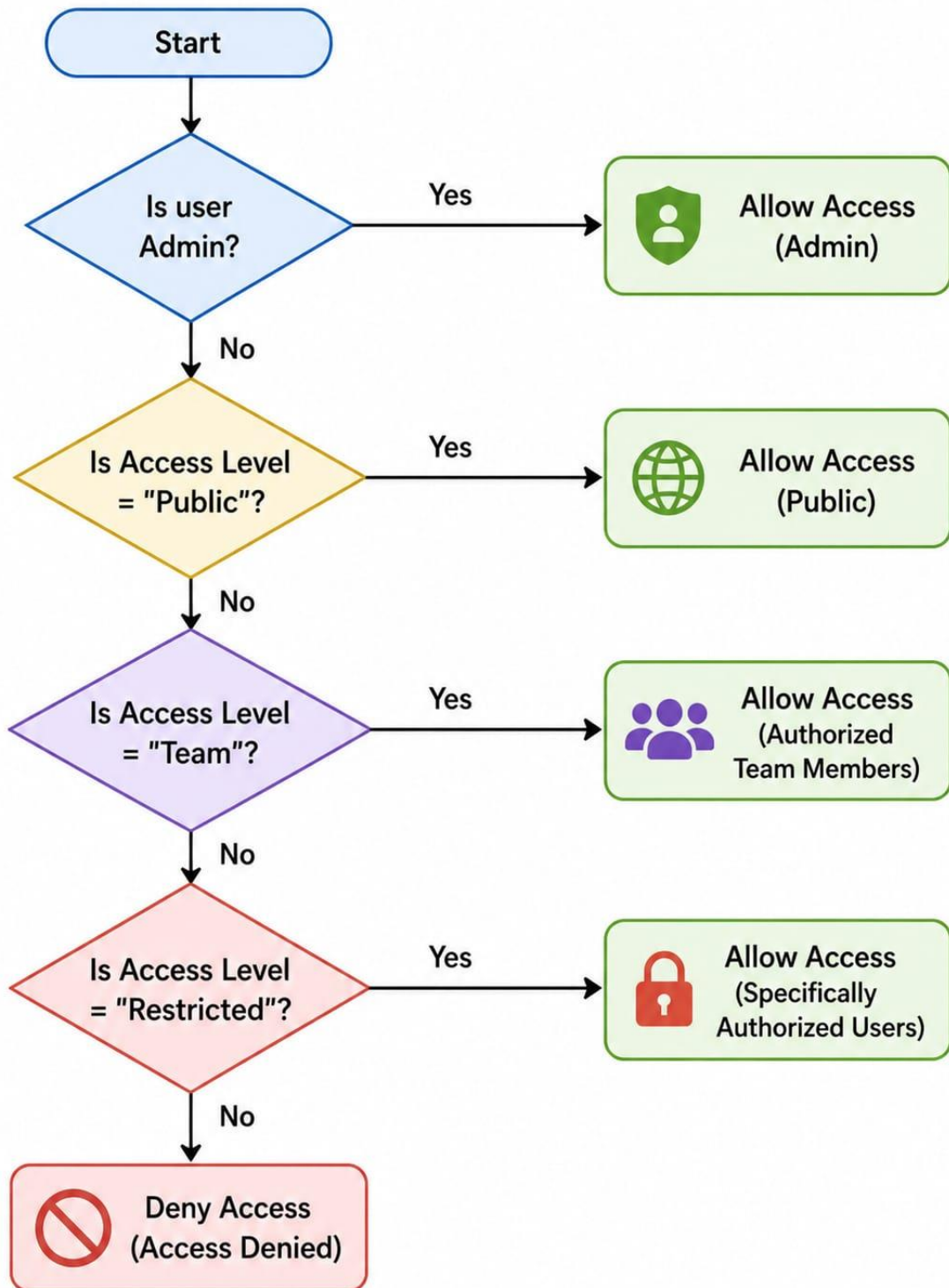
1. Read
2. Write
3. Create
4. Delete

For example, a Read ACL can be configured to allow access only when the user satisfies the required conditions.

4. Implement Script-Controlled Access

The ACL script checks the field value before allowing access.

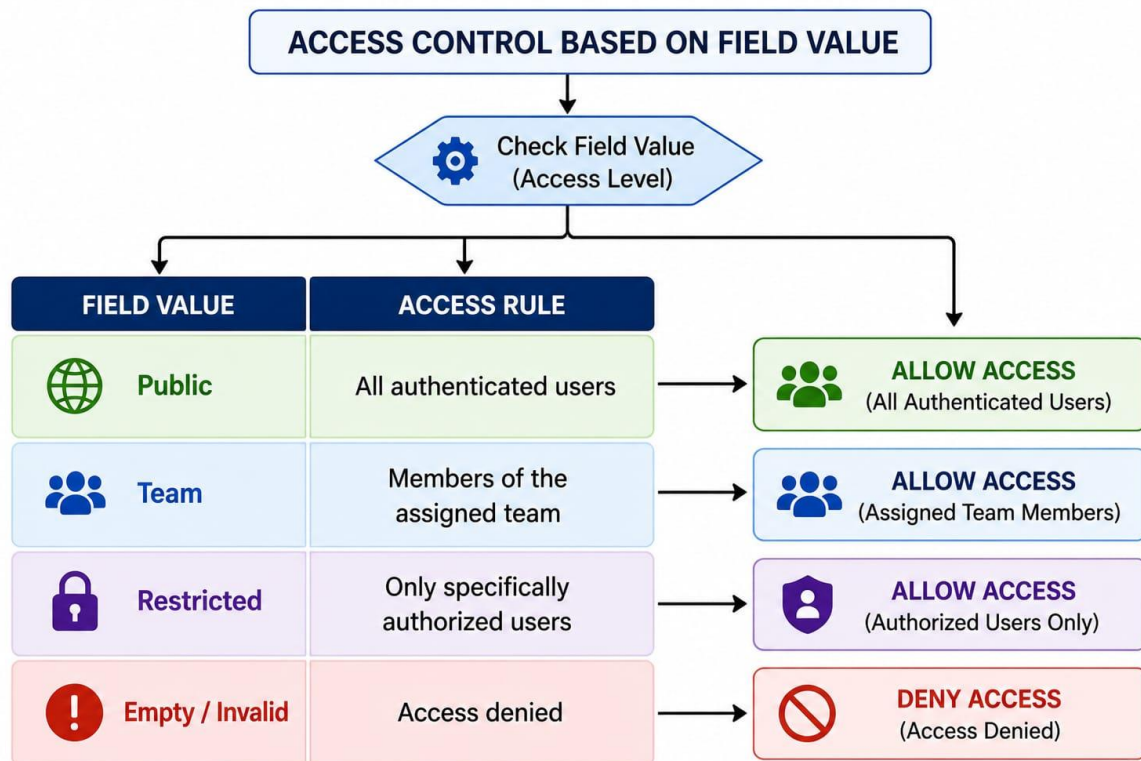
Conceptual logic:



5. Field Value-Based Restriction

The important part of the project is that access is dynamically controlled according to the field value.

For example:



This prevents users from accessing sensitive records simply because they have general access to the table.

6. Script Validation

The ACL script verifies:

- Current user's identity
- User's role
- Field value
- Team or group membership
- Authorization status

The result is then used to allow or deny access.

7. Testing During Development

Different scenarios are tested:

Test 1: Authorized user + permitted field value → Access Granted

Test 2: Unauthorized user + restricted field value → Access Denied

Test 3: Admin + any field value → Access Granted, according to the project's admin rule.

Test 4: User changes the field value → ACL evaluates the new value and applies the corresponding access rule.

Development Outcome

The final system provides dynamic, field-level security through scripted ACLs. Instead of giving every user the same access, the system evaluates the record's field value and the user's authorization before permitting the requested operation.